

Critical tech for critical business needs

Intel's new E7 servers are high on tech, geared for business, meant for mission critical services that businesses can't do without

Jayesh Shinde

jayesh.shinde@digit.in

In the age of the Internet, where all of us are hooked onto content consumption portals, indulging in ecommerce, and uploading humongous amounts of data in the cloud, powering the backend of all the services we love is a herculean exercise. Data center farms with servers processing millions of instructions per second, crunching petabytes and zetabytes of data. And where traditional CPUs have dual-core, quad-core configurations, imagine a processor with 18 dual-threaded cores. Quite a powerhouse, which is what the new Intel Xeon E7 v3 chips are essentially. With these new server chips, Intel has taken x86 computing to new heights, beating traditional POWER8 and RISC-based servers in over 20 new performance benchmarks.

What's new?

Aimed at higher end of the market meant for mission critical computing needs, the sparkling new Xeon E7 v3 Haswell processor comes with a number of improvements over its Ivy Bridge predecessor (the E7 v2), which include increased number of cores, increased cache size and support for the latest DDR4 memory which has faster speeds and lower power consumption than the older DDR3.

Intel has made a number of small but significant improvements to the processor which include separate clock and voltage domains for each core which make the

already power efficient processor even more optimized by allocating power only where it is actually needed. Also added is the new multiple rank sparing and DDR4 addr/cmd parity error recovery which has led to an improved uptime, increased system robustness and reduces the overall service costs. Not only is the number of cores increased but also the performance per core has been significantly improved. This has been possible because of the

5-10 times better performance per dollar compared to traditional IBM Power-based systems. Go here, for more info: <http://www.intel.com/content/www/us/en/processors/xeon/xeon-e7-details.html>

Who needs these chips?

Any business that's dependent on big data analytics on a real-time basis, sensitive to scaling needs, develop unique insights in your business, drive customers, and create new business models to transform organizational processes should look for servers running on Intel's new Xeon E7 v3 processors. According to Intel, seventeen server manufacturers are slated to start shipping

Features	Intel® Xeon® processor E7-8800/4800/2800 v2 product family (IVB-EX)	Intel® Xeon® processor E7-8800/4800 v3 product family (HSW-EX)
Socket	R1	
Process technology	22 nm	
Cores	Up to 15	Up to 18
TDP	155W Max	165W Max
AVX Support	AVX 1 - 8 DP Flops/Clock/Core	AVX 2 - 16 DP Flops/Clock/Core
Intel® QPI Ports/speed	3x Intel QPI v1.1, 8.0 GT/s max.	3x Intel QPI v1.1, 9.6 GT/s max.
Core Addressability	46 bit / 48 bit virtual	
Last Level Cache Size	Up to 37.5MB	Up to 45MB
Memory DDR4 speeds	N/A	Perf Mode: 1333 MT/s, 1600 MT/s Lockstep Mode: 1333, 1600, 1866 MT/s
Memory DDR3 speeds	Perf Mode: 1066, 1333 MT/s Lockstep Mode: 1066, 1333, 1600 MT/s	Perf Mode: 1066, 1333 MT/s, 1600 MT/s Lockstep Mode: 1066, 1333, 1600 MT/s
VMSE Speeds	Up to 2667 MT/s	Up to 3200 MT/s
DIMMs/Socket	24 DIMMs/(3 DIMMs/DDR3 channel)	24 DIMMs/(3 DIMMs/DDR4 or DDR3 channel)
RAS	WSM EX Baseline Features + • eMCA Gen 1 • MCA recovery - Execution Path • MCA IO • PCIe LER	IVB-EX Baseline + • eMCA Gen2 • Address Based Memory Mirroring • Multiple Rank Sparing • DDR4 recovery for command/address parity errors
Intel® Integrated I/O	32 PCIe* 3.0, 1x x4 DMI2	
Security	Intel® TXT, Intel® AES-NI, Intel® Platform Protection with OS Guard, Intel® Data Protection with Secure Key	
Common Platform Technologies	Intel® Run Sure Technology, Intel® Turbo Boost Technology, Intel® Hyper-Threading Technology, Intel® Dynamic Power, Intel® VT-x, Intel® VT-d, Intel® VT-c, Intel® I/OAT/CB3, Intel® Node Manager, Intel® AVX	

addition of two fully buffered rings, which have increased the bandwidth per core, and TSX, that not only allows transactional execution without locking but also Fine grain behavior at coarse grain effort. The v3 also comes with integrated IO hub that has increased both the PCIe bandwidth and latency tolerance, and also has better virtualization capabilities thanks to the adaptive LLC allocation policy and the Cache allocation monitoring. What does all of this translate to? Performance boost of 40 per cent compared to Xeon E7 v2, and

Xeon E7 v3 systems later this year, and especially with a large server refresh cycle taking place in 2015, that should present lots of businesses an opportunity to make use of Intel's offerings. For IT managers and hardware technology vendors who are cost-conscious, sure these new Intel Xeon E7 v3 chips aren't available for cheap. But if time is money, and depending on your unique business needs, it may be a worthwhile investment to consider. If there's one thing about silicon in the realm of computing - newer is almost always better. **d**